

Rishab Vempati

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Education

Purdue University

B.S. in Computer Science & Artificial Intelligence, GPA: 3.8/4.0

West Lafayette, IN

Expected Graduation: Dec. 2027

Coursework: Algorithms, Data Structures, Operating Systems, Computer Architecture, Databases, Artificial Intelligence

Experience

IBM

Aug. 2026 – Present

Software Engineering Intern

Austin, TX

- Building a deployment intelligence platform unifying CI/CD, Git, container, log, metric, trace, and monitoring data.
- Developing ML root-cause ranking that correlates deployment changes with failures across observability signals.
- Training anomaly-detection models to flag abnormal deployments and prioritize likely causes for incident triage.

CIDER Lab

Feb. 2026 – Present

Undergraduate Researcher

West Lafayette, IN

- Built distributed evaluation infrastructure executing **21K+** computer-use-agent rollouts across **879 websites** on H100s.
- Co-authored uxCUA (arXiv:2604.26020), training VLM-based computer-use agents with reinforcement learning.
- Researching UI generation from implicit feedback using interaction traces, dwell time, and task-completion signals.

Cora

Jan. 2026 – Jul. 2026

Founding Engineer

San Francisco, CA

- First engineer on a 10-person team; owned Cora's codebase end-to-end from prototype to first enterprise client.
- Built live meeting and email ingestion, chunking diarized transcript streams into retrieval-ready context for live Q&A.
- Shipped to a **500+ seat** enterprise client; owned deployment, observability, on-call, and incident response.

Ekai

Aug. 2025 – Dec. 2025

Software Engineer (Part-Time)

Remote

- Built vector and SQL retrieval over **10K+ documents** with **92% retrieval accuracy** for enterprise knowledge search.
- Deployed retrieval through an MCP server, enabling shared context and tool-accessible knowledge across AI-agent clients.
- Added response caching and query optimizations, cutting p95 latency from **8.2s to 1.8s** for **50+ concurrent users**.

AssetMark

Jun. 2025 – Aug. 2025

Software Engineering Intern

San Francisco, CA

- Migrated a legacy .NET/C# communication service to event-driven microservices backed by Azure Service Bus.
- Implemented dead-letter queues and exponential-backoff retries, reducing message failures from **0.8% to 0.12%**.
- Enabled autoscaling for **1M+ quarterly emails** while sustaining **99.7% delivery success** during traffic spikes.

Cisco

Jan. 2024 – May 2024

Undergraduate Researcher

West Lafayette, IN

- Engineered an automated forecasting-model selection pipeline using Bayesian inference across **10K+ product SKUs**.
- Built Isolation Forest and Gaussian Mixture Model anomaly-detection pipelines, improving forecast accuracy by **3%**.

Projects

Computer-Use Agent Evaluation Platform | [Code](#) | Python, BrowserGym, Modal, vLLM

- Built a reproducible harness comparing generalist and specialized browser agents across task families, models, and seeds.
- Specialized agents improved task success from **64% to 100%** while cutting mean latency **46%** (53.6s to 28.7s).
- Reduced invalid actions from **10% to 0%**; added resumable execution, hard budgets, telemetry, and routing baselines.

Latent Communication in Language-Model Agents | Python, PyTorch, Transformers

- Audited hidden-state communication between Llama-3.1-8B-Instruct agents through causal activation interventions.
- Found decodable semantics could fail behaviorally; receiver-preserving edits raised transfer from **1/64 to 32/64**.

Technical Skills

Languages: Python, Java, C++, C, SQL, JavaScript/TypeScript, Bash

AI/ML: PyTorch, Hugging Face Transformers, vLLM, BrowserGym, MCP, SLURM

Systems: Linux, Docker, Kubernetes, Kafka, Redis, PostgreSQL, FastAPI, gRPC/REST, CI/CD, AWS, Azure, Grafana